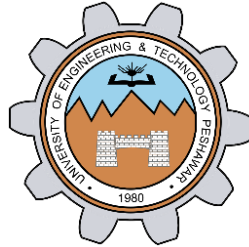


Serial Communication

LAB # 13



Spring 2024

CSE-307L Microprocessor Based System Design Lab

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Class Section: **C**

“On my honor, as student of University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work.”

Submitted to:

Dr. Asif Ali Khan

Date:

1st June 2024

Department of Computer Systems Engineering
University of Engineering and Technology, Peshawar

Task:

Code:

Transmitter Code:

```
lab_task.c
1  #include<reg51.h>
2
3  sbit intr = P3^2;
4  sbit rd = P3^4;
5  sbit wr = P3^5;
6  int i, temp;
7
8  void delay(int n){
9      for(i = 0; i< n; i++){
10         }
11     }
12
13 void ADC(){
14     rd = 1;
15     wr = 0;
16     delay(10);
17     wr = 1;
18     while(intr == 1);
19     rd = 0;
20     delay(10);
21 }
22
23
24 void serial_comm(){
25     TMOD = 0x20;
26     TH1 = 0xFD;
27     SCON = 0x50;
28     PCON = 0x00;
29
30     TR1 = 1;
31 }
32
33 void main(){
34     serial_comm();
35     while(1){
36         ADC();
37         SBUF = P1;
38         while(TI == 0);
39         TI = 0;
40     }
41 }
42
43
```

Receiver Code:

```
task2.c
1  #include<reg51.h>
2  int i;
3  void delay(int n){
4      for(i = 0; i< n; i++){
5          }
6      }
7
8  void serial_comm(){
9      TMOD = 0x20;
10     TH1 = 0xFD;
11     SCON = 0x50;
12     PCON = 0x00;
13     TR1 = 1;
14 }
15
16 void main(){
17     serial_comm();
18
19     while(1){
20         while(RI == 0){
21             RI = 0;
22             P1 = SBUF;
23         }
24     }
25 }
```

Proteus Simulation:

